Ogbonna Arinze

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SUMMARY

Full-Stack Engineer with 8 years of professional experience building high-impact web applications from the ground up across healthcare, fintech, civic tech, and enterprise environments. Proven ability to ship 0-to-1 products end-to-end using **TypeScript**, **Next.js**, **React**, **NestJS**, **PostgreSQL**, **Prisma**, and **Docker**, with deep AI integration into both products and daily development workflows. Experienced in making pragmatic architecture decisions with high autonomy, building production AI/RAG systems, and mentoring engineers, all while iterating fast based on real user feedback.

CORE COMPETENCIES

- **Full-Stack Development:** TypeScript, Next.js, React, NestJS, NodeJS, REST APIs, SSR/SSG, and monorepos (Nx, pnpm workspaces).
- **Databases & ORM:** PostgreSQL (pgvector), MongoDB, Prisma, and database schema design.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, and multi-agent workflows.
- **Cloud & Infrastructure:** Docker, Docker Compose, Vercel, AWS (CloudFront, S3), GitHub Actions, CI/CD pipelines, and Traefik.
- **Frontend & UI:** React, Tailwind CSS, Radix UI, Recharts, Ant Design, Storybook, design systems, and performance optimization.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Leadership:** Technical architecture decisions, mentorship, Agile/Scrum, product-minded development, and 0-to-1 product ownership.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, optimizing workflows for clinical site integrations and enhancing adoption.
- Architected a reusable **design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing lazy loading, code splitting, dynamic imports, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies — Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS**, and **Axios**.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered civic tech platform for Nigerian budget analysis and corruption tracking. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected and built the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a hybrid **RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, Reciprocal Rank Fusion, and **Cohere** reranking.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, processing 700+ documents (959 files) across 37 states with zero ingestion errors.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts** (22 interactive chart types), featuring dark mode, URL state management, and responsive mobile layouts.
- Configured CI/CD with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, phone OTP authentication, Telegram bot with webhook handling, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*